SAAS CUSTOMER CHURN INTELLIGENCE

PROJECT REPORT | Exploratory analysis of customer churn patterns

50,000	916	1.83%	911
records analyzed	churned records	overall churn share	unique churned companies

Objective & approach

This project explores SaaS customer churn using Python, pandas, Matplotlib and Seaborn. The notebook standardizes column names, checks data quality and summary statistics, converts date fields, isolates churned records, and compares churn counts across industry, plan tier, payment status, country, account manager and time. The dataset contains 50,000 rows and 18 columns.

Key findings

Industry	Among the five leading industries reported, EdTech had the highest churn count (128), followed by DevTools (121) and Logistics (119). CyberSecurity was lowest among these five (112).
Plan tier	Starter accounted for 685 churned records, compared with Pro (174) and Enterprise (57), making it the largest plan-tier group in the churned subset.
Payment status	Active: 513 (56.0%); Failed: 294 (32.1%); Pending: 109 (11.9%). Failed payments are a notable area for further investigation.
Geography & timing	Singapore had the largest country count (135), while the UK had the smallest (95). By last-active year, 742 churned records were in 2025 and 174 in 2026; December was the most frequent month (212).

Recommendations

Prioritize retention diagnostics for the high-volume Starter segment and EdTech customers; review failed and pending payment journeys; and compare churn rates (not just counts) within each industry, plan tier and country. Segmenting by tenure, feature adoption and last activity may help reveal actionable early-warning signals. These are exploratory findings: counts alone do not establish causes of churn.

PREPARED BY

Osayi Chukwurimma Victoria Phone: 08114569423 Email: rimmavictoria82@gmail.com

Source: “SAAS CHURN.ipynb” uploaded project notebook. Churn share calculated as 916 / 50,000. Industry, plan-tier, payment, country and time figures are reproduced from the notebook outputs.